

Exercise 1

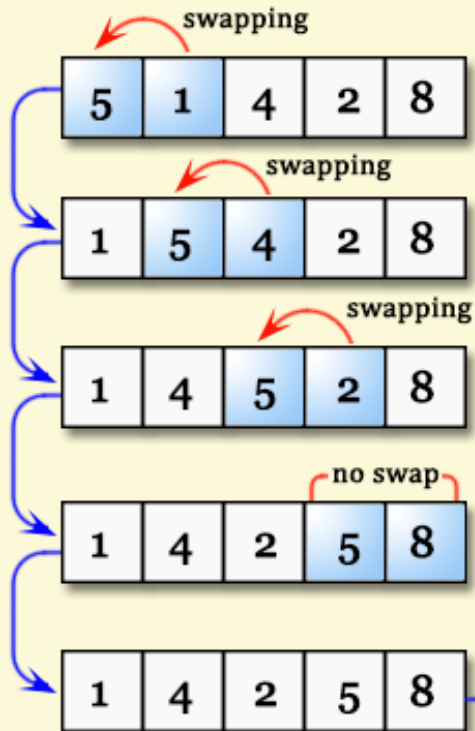
To sort 5, 1, 4, 2, 8 using Bubble Sort

- Please write the swaps and the sorting result of each pass by bubble sort

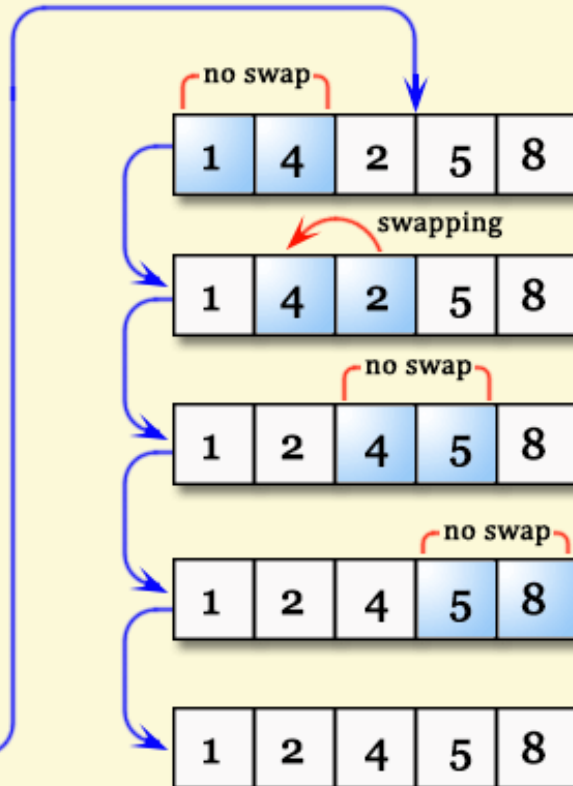
Exercise 1

To sort 5, 1, 4, 2, 8 using Bubble Sort

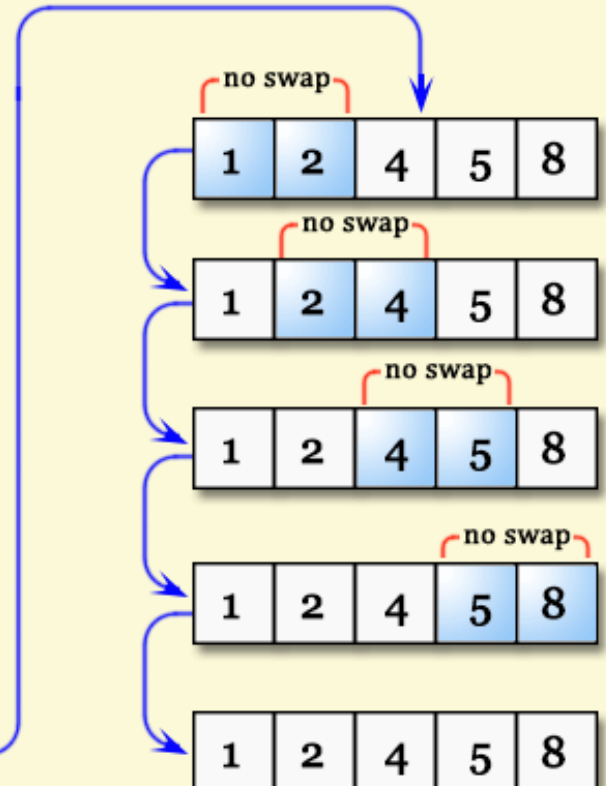
First Pass



Second Pass



Third Pass



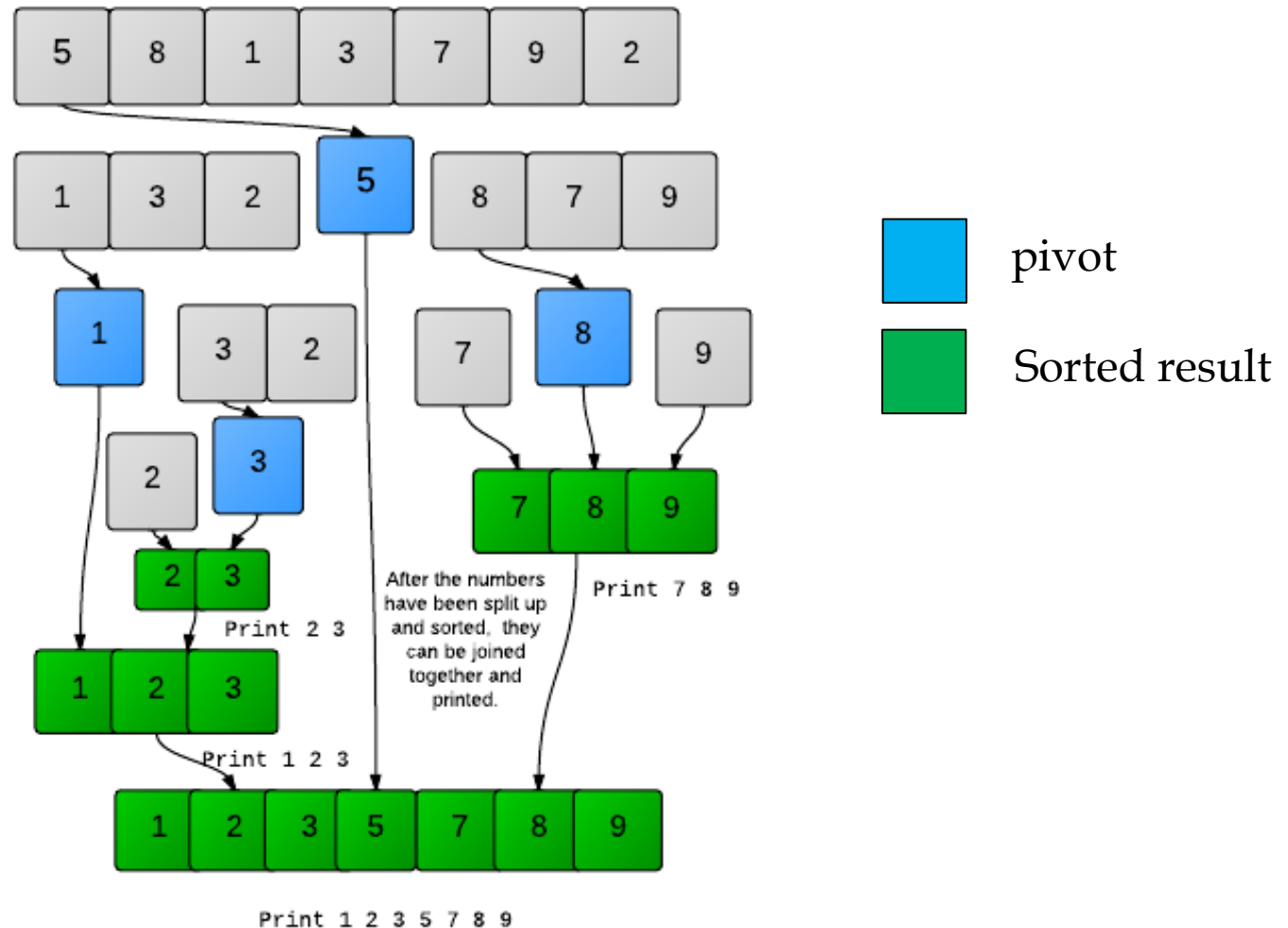
Exercise 2

To sort 5, 8, 1, 3, 7, 9, 2 using Quick Sort

- Please select the first element as the pivot in each partition
- Please write the pivot and sorting result of each iteration by quick sort

Exercise 2

To sort 5, 8, 1, 3, 7, 9, 2 using Quick Sort



Exercise 3

- Give asymptotic upper bounds (Big-O notation) for $T(n)$. Make your bounds as tight as possible.

$$T(1) = 1$$

$$T(n) = T(n/2) + n \quad \text{if } n > 1$$

Exercise 3

$$T(1) = 1$$

$$T(n) = T(n/2) + n \quad \text{if } n > 1$$

Set $h = \log_2 n$

$$T(n) = n + T(n/2)$$

$$= n + n/2 + T(n/2^2)$$

$$= n + n/2 + n/2^2 + T(n/2^3)$$

...

$$= n + n/2 + n/2^2 + \dots + n/2^{h-2} + n/2^{h-1} + T(n/2^h)$$

$$= n(1 + 1/2 + 1/2^2 + \dots + 1/2^{h-2} + 1/2^{h-1}) + T(n/2^h)$$

$$\leq n(1 + 1/2 + 1/2^2 + \dots + 1/2^{h-1} + \dots) + T(n/2^h)$$

$$= 2 \cdot n + T(1)$$

$$T(n) = O(n)$$

Exercise 4

- Give asymptotic upper bounds (Big-O notation) for $T(n)$. Make your bounds as tight as possible.

$$T(1) = 1$$

$$T(n) = 4T(n/2) + n \quad \text{if } n > 1$$

Exercise 4

$$T(1) = 1$$

$$T(n) = 4T(n/2) + n \quad \text{if } n > 1$$

Set $h = \log_2 n$

$$T(n) = n + 4T(n/2)$$

$$= n + 2n + 4^2 T(n/2^2)$$

$$= n + 2n + 2^2 n + 4^3 T(n/2^3)$$

...

$$= n + 2n + 2^2 n + \dots + 2^{h-2} n + 2^{h-1} n + 4^h T(n/2^h)$$

$$= n(1 + 2 + 2^2 + \dots + 2^{h-1}) + 4^h T(n/2^h)$$

$$= n \frac{2^h - 1}{2 - 1} + 4^h T(n/2^h)$$

$$T(n) = n(n - 1) + n^2 T(1) = O(n^2)$$